

Lab: Case Study Analysis — Planning, Decision-Making, and Their Disorders

Objective

Analyze clinical and experimental cases illustrating the neural basis of planning, applying Active Inference's policy evaluation framework.

Part 1: Frontal Lobe Syndrome

Case: Patient P.G., 55, suffered a large bilateral prefrontal cortex lesion from a meningioma. Post-surgery, P.G.'s IQ and language are intact, but he cannot organize his daily life. He cannot plan a meal (starting to cook without checking for ingredients), dresses in random order, and shows "utilization behavior" — picking up and using any object placed in front of him (a comb, a toothbrush, a pencil) without purpose.

- How does loss of PFC impair hierarchical policy evaluation while leaving basic motor control intact?
- Why does P.G. show utilization behavior? (Hint: without hierarchical policy evaluation, the most immediately available policy is executed.)
- How does this case illustrate the difference between knowing *how* to perform an action and knowing *when* and *why* to perform it?

Write your response here

Part 2: Impulsivity and Temporal Discounting

Case: A behavioral economics study offers participants the choice between \$10 now and \$20 in two weeks. Participants with higher trait impulsivity (measured by the Barratt

Impulsiveness Scale) consistently choose the immediate \$10, even when the two-week delay is trivial relative to their financial needs.

- Interpret temporal discounting in Active Inference terms: how does low precision on long-horizon policies bias selection toward immediate rewards?
- What does this suggest about the "planning horizon" encoded in the impulsive individual's generative model?
- How does dopamine modulate this temporal horizon? (Consider the effect of stimulant medication on temporal discounting in ADHD.)

Write your response here

Part 3: Depression and Behavioral Paralysis

Case: Patient L.K., 42, with major depressive disorder, describes: "I know I should go for a walk, call my friend, apply for the job. But what's the point? Nothing will change. Everything will go wrong." She spends most of her time in bed, not from lack of energy but from a conviction that no action will lead to a positive outcome.

- How does Active Inference model depression as high precision on negative outcome predictions across all policies?
- Why does this produce behavioral inertia? (Hint: if all policies lead to high expected free energy, none is clearly better than inaction.)
- What does this framework suggest about therapeutic approaches? (Consider interventions that increase precision on positive predictions — behavioral activation, cognitive restructuring.)

Write your response here

Part 4: Addiction and Policy Narrowing

Case: Patient R.S., 38, with opioid use disorder, reports: "I know the drugs are destroying my life. I've lost my job, my family. But when the craving hits, it's like nothing else matters. My whole world shrinks to one thing." R.S. has attempted abstinence multiple times but relapses within 2-3 weeks.

- How does Active Inference model addiction as progressive narrowing of the policy space?
- What role does dopaminergic sensitization play in increasing precision on drug-seeking policies?
- How does this framework explain relapse even after months of abstinence? (Hint: the generative model's structure has been reshaped; precision on the drug-seeking policy remains high under certain cues.)

Write your response here

Part 5: Synthesis

Write a 200-word synthesis explaining how all four cases demonstrate that planning is hierarchical policy evaluation — and that specific disruptions of this hierarchy produce distinct clinical presentations.

Write your response here

Lab Summary

Part	Skill Developed	Case Type
1	Neuropsychological reasoning	Frontal lobe syndrome
2	Behavioral economics analysis	Impulsivity and temporal discounting
3	Clinical reasoning	Depression and planning paralysis
4	Addiction analysis	Opioid use disorder and policy narrowing
5	Integrative synthesis	Connecting planning disorders to hierarchical inference